DISSERTATION II REPORT

Submitted in partial fulfilment of the requirements for the degree of

M.Tech. Computer Science and Engineering

(Information Security)

by

Lokesh S

24MCI0029

Under the Supervision of

Dr. Manikandan K

Professor Grade 1

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

March, 2026

DECLARATION

I, **Lokesh S (24MCI0029)** hereby declare that the thesis entitled **"Block Chain - Assisted Secure Document Sharing System"** submitted to Vellore Institute of Technology (VIT), Vellore for the award of the degree of **Master of Technology in Computer Science and Engineering (Information Security)** is a record of bonafide work carried out by me under the supervision of **Dr. Maniknadan K**, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

Date: March 13, 2026

Signature of the Candidate

CERTIFICATE

This is to certify that the thesis entitled **"Block Chain - Assisted Secure Document Sharing System "** submitted by **Lokesh S (24MCI0029)**, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore for the award of the degree of **Master of Technology in Computer Science and Engineering (Information Security)**, is a record of bonafide work carried out by him/her under my supervision during the period, **10-12-2025** to **20-04-2026** as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other Institute or University. The dissertation fulfills the requirements and regulations of the Institute and in my opinion meets the necessary standards for submission.

Place: Vellore, signature of the guide

Date: March 13, 2026 Dr. Manikandan K

Internal Examiner **External Examiner**

Head of the Department

Department of Computer Science and Engineering

EXECUTIVE SUMMARY

In fact, with the fast digitalization of sensitive documents, such as medical reports, forensic evidence, legal documents, etc., a multitude of serious issues concerning data integrity, unlawful amendments, and the lack of trust have been hurled into the arena when organizations and agencies are exchanging information. The old fashioned centralized document management systems simply do not pass the test in applications like healthcare and law where high stakes are involved due to being simple to tamper with, single points of failure, and because their audit trails are highly opaque.

Therefore, I am suggesting a Secure Document Sharing System with crypto-assistance based on a blockchain and multi-chain that would encrypt, imprison, and verifiably verify the confidentiality, integrity, and integrity of sensitive forensic data. In essence, a distinct SHA-256 hash is assigned to each document thus you can identify any unlawful manipulation in a blink of an eye. At that point, we e-authenticate reports and repudiate them using RSA-based digital signatures. The sidechain hashes are anchored by a main chain to ensure the entire world is unchangeable and irreversible, whilst transaction hashes are stored by the sidechain of things such as medical reports, recorded evidence and verification records.

The system has a role-based access control that involves all the key actors such as the doctors, reviewers, and court officials. Once the liveness checks are passed, the docs are able to send case reports and digitally sign evidence and use those signatures and hash to prove authenticity in the court. Authorities can confirm the authenticity of evidence by merely giving the case ID or hash key, but not viewing the actual private data.

Overall, this demonstrates that web-based forensic systems can be connected to blockchain technology to provide a safe, transparent, and legally viable paradigm of digital evidence. It also provides an expandable and impregnable infrastructure which is used in verification of forensic and judicial documentation and enhances confidence, accountability and integrity of data.

Keywords: *Blockchain, SHA-256 Hashing, Data Integrity, Main Chain and Side Chain Architecture, Tamper-Proof Records, Forensic Evidence Management, Court Verification System.*

ACKNOWLEDGEMENTS

With immense pleasure and a deep sense of gratitude, I wish to express my sincere thanks to my supervisor **Dr. Manikandan K**, School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Vellore without his/her motivation and continuous encouragement, this research would not have been successfully completed.

I am grateful to the Chancellor of VIT, **Dr. G. Viswanathan**, the Vice Presidents, and the Vice Chancellor for motivating me to carry out research in the Vellore Institute of Technology.

It would be no exaggeration to say that the Dean in-charge of SCOPE, **Prof. Jaisankar N**, was always available to clarify any queries and clear the doubts I had during the course of my project. I would also like to acknowledge the role of the HoD, **Dr. Gopinath M.P**, who was instrumental in keeping me updated with all necessary formalities and posting all the required formats and document templates through the mail, which I was glad to have had.

Finally, I would like to thank **Vellore Institute of Technology** for providing me with infrastructural facilities, a flexible choice and for supporting my research and execution related to the dissertation work.

Place: Vellore, **Lokesh S**

Date: March 13, 2026 **(24MCI0029)**

CONTENTS

1 INTRODUCTION **1**

1.1 Overview 1

1.2 Objectives 2

2 LITERATURE REVIEW **4**

2.1 Review of Literature 4

2.2 Review Types 5

2.2.1 Process and Product 6

2.3 Page Limit of Review 7

3 PROBLEM DESCRIPTION **10**

3.1 Problem Description 10

3.1.1 Dataset Description 11

4 PROPOSED APPROACH **13**

4.1 Overview of the Proposed System 13

4.2 System Architecture 14

4.2.1 User Layer 14

4.2.2 Application Layer 15

4.2.3 Cryptographic Layer 15

4.2.4 Off-Chain Storage Layer 16

4.2.5 Sidechain Layer 16

4.2.6 Main Blockchain Layer (Anchor) 16

4.2.7 Verification Layer (Court Validation System) 17

4.3 Algorithms 17

4.3.1 Secure Hash Algorithm 17

4.3.2 Rivest–Shamir–Adleman. 18

4.3.3 Side Chain 18

4.3.4 Main Chain 18
4.4 Proposed Model Implementation 19
4.4.1 Acquisition of Evidence and Generation of Hashes. 19
4.4.2 RSA based Digital Signature Generation. 19
4.4.3 Side Chain Evidence Storage 19
4.4.4 Main Chain Anchoring 19
4.4.5 Evidence Verifying Process. 20
4.5 Training Strategy 20
4.5.1 Hyperparameter 20
4.6 Advantages of the Proposed Approach 21
4.6.1 Data Integrity Assurance 21
4.6.2 Strong Authentication and Non-Repudiation 21
4.6.3 Tamper-Resistant Storage 21
4.6.4 Improved Scalability with Side Chain Architecture 21
4.6.5 Enhanced Security through Main Chain Anchoring 21
4.6.6 Transparency and Traceability 22
4.6.7 Reliable Evidence Verification 22

5 EXPERIMENTAL SETUP 23
5.1 Hardware Setup 23
5.2 Software Setup 23
5.3 Dataset Description 24
5.4 Training Procedure 24

6 RESULT ANALYSIS AND DISCUSSION 25
6.1 Performance Evaluation 25
6.2 Visualization of Results 27
6.2.1 Visualization of Hash Key 27
6.2.2 Visualization of Court-View and Block Chain Model 28
6.3 Comparative Analysis 28
6.4 Interpretation of Findings 29
6.5 Limitations and Insights 29

7 CONCLUSIONS AND FUTURE ENHANCEMENTS 30
7.1 Summary of the Research 30
7.2 Conclusions Drawn from the Study 30

7.3 Importance of the Research 30
7.4 Recommendations for Future Research 31
7.5 Recommendations for Practitioners 31
7.6 Final Summary 31

LIST OF FIGURES

4.1 Architecture Diagram 14

6.1 Confusion Matrix 25

6.2 ROC Curve 26

6.3 Bar Graph 26

6.4 Case Detail and Hash Key 27

6.5 Verification and Block Chain 28

LIST OF TABLES

4.1 Hyperparameter Configuration of the Proposed System 20

List of Abbreviations

RSA Rivest Shamir Adleman

SHA Secure Hash Algorithm

List of Symbols

B_i Current block hash

B_n Last block in the sidechain

d Private key

D_i Document hash stored in the block

$H(\cdot)$ SHA-256 hash function

H_{SC} Latest sidechain hash

M Input message

M_i Current main chain block hash

S Digital signature

SC_i Sidechain hash

T_i Timestamp

Chapter 1

INTRODUCTION

1.1 Overview

In the present times, it is extremely important to safeguard digital information within the legal, forensic, and health care sectors. Every person who has ever had to handle medical reports, forensic evidence, or court documents understands that one should have a good security against unauthorized access, manipulation, and data loss. Due to the fact that most systems continue to operate on centralized servers that are dependent on a small number of parties, they are susceptible to single-point failure and manipulation and attacks.

This is where blockchain enters the picture as a trendy solution to the necessity of accountability, transparency and data integrity. Having a decentralized and immutable list of transactions, blockchain ensures the authenticity and traceability of any document remain adhered to. It is possible to check the integrity of a file without exposing its contents, owing to hash functions such as SHA 256.

In the project, named, Blockchain-Assisted Secure Document Sharing System, I suggest a multi-chain architecture. It has one chain and a number of side chains such that after a document is created, validated, and recorded it cannot be covertly changed. Role access controls ensure that the system is only accessed by entitled individuals, such as doctors, reviewers and court officials.

1.1 Outline of Thesis

The thesis provides a full-scale framework of the development of a secure, impenetrable and verifiable document-management platform based on blockchain. The abstract provides a brief overview of the motivation, method, system design and key findings. The introduction plunges into the issues of centralized storage- manipulation, non transparency, unreliable verification in the legal and medical environment.

The issue statement emphasizes the importance of trustful and immutable digital documents to legal and medical processes, and the goals are safe exchange, integrity, cryptographic validation, and role-based controls. A literature survey verifies the existing approaches: centralized databases, encryption-only solutions, and initial blockchain storage frameworks and identifies their weaknesses in terms of scalability, privacy, and real-time authentication.

The suggested workflow is described: generate the document and calculate a SHA -256 hash, write the output to a side chain and anchor the side-chain hash to the main chain to ensure worldwide immutability. The architecture section describes the concept of multi-chain, front-end, back-end, and cryptographic tool that ensure the security of communication. Implementation and methodology include data format and hashing, verification logic and system integration.

The findings and discussion chapter will reveal how hash comparison and blockchain traversal can confirm the data integrity, tamper detection and secure verification, which demonstrates that the system is much more reliable than traditional ones. Lastly, there is the conclusion proposing improvements in the future with biometric authentication, distributed storage, and a massive implementation to use in forensics and legal. It also shows the contributions of the project and proves the strength of the proposed framework. Those references and appendices provide additional reading, the description of the algorithm, and snippets of the code and screenshots of the interface.

1.2 Objectives

The aim of the Blockchain-Assisted Secure Document Sharing System is to ensure the integrity of data, its authenticity, and verifiable transparency of sensitive records with the combination of blockchain, cryptographic hashing, and role-based access. The system can handle cases of tamper-proof processing in areas such as healthcare, forensics, and court cases where such data loss or manipulation can have severe effects.

The system generates unique hash keys that provide the actual state of each record through SHA 256. These hash numbers are stored on a side chain which performs document specific transactions- report submission, verification approval and multimedia evidence storage. The last side-(chain)-hash is anchored to the primary blockchain to enhance the immutability and international trust, any unlawful manipulation can be immediately detected through comparing hash and hopping up the chain.

The multi-stakeholder workflow enables authorized parties to upload, verify, approve, and validate documents without viewing the original data, which are, in turn, posted in a safe way

by using case identifiers or hash keys, and, crucially, physicians, reviewers, and court officials can be considered authorized parties. This will allow a trusted third party to check the data, but still keep it confidential. Separating document storage to blockchain anchoring assists in scaling and maintaining privacy, which lowers the storage expenses whilst maintaining cryptographic assurances.

Offering a verifiable, immutable record of the entire document operations will create trust among stakeholders, as well as enable digital records to be produced in a court of law. The eventual goal is to develop a privacy-sensitive, scalable, and legally defensible system of evidence management that enhances responsibility, reduces chances of manipulation, and gives it a strong basis to be used in practical health-related and legal context.

Chapter 2

LITERATURE REVIEW

2.1 Review of Literature

Issues of support of data integrity, confidentiality, and trust in sharing documents have increased due to the rising amount of sensitive information digitized in law, forensics and health. In the earlier years, the majority of digital doc-management systems were centralized and based on the outdated databases and access control. They were quicker and more efficient to store, however, it was extremely easy to be hacked by insiders, they could illegally manipulate data, hit a single point of failure and the audit trails were super opaque. Everything that rendered data validity and legal admissibility weak, particularly when one is working with medical and court records. People escaped such limitations by implementing crypto techniques such as hashing and encryption. hashing (MD5, SHA-1, then SHA-256) provided us with fixed-size digests of the content, increasing the ability to detect tampering. But the gotcha? They continued to rely on centralized storage and relied on third-party trust and so they were not able to provide the immutability, decentralized trust, or independent verification required of an environment such as a forensic laboratory, a court or a hospital.

The second major innovation was blockchain. The blockchain had ensured reputation and transparency through decentralized ledgers, cryptographic block connections, and consensus. It was tested to safeguard the digital evidence, legal documents and medical records. However, due to the fact that blockchains are usually public, the earliest attempts were to upload entire documents on the blockchain, which scaled badly, killed storage costs, and raised privacy concerns.

That is why researchers proposed one more variant to use a hybrid solution: leave the majority of the information off-chain in safe databases or distributed storage and transfer hashes to the chain. This lightened the storage expenses and was more effective at ensuring privacy. However, a good number of present configurations do not give the flexibility to support industry-specific workflows - consider medical validation, forensic validation, court authentication - despite the

hybrid deal consuming less space.

In an effort to increase scalability and modularity, more recent work considers sidechains and multi-chain designs. The core chain remains the trust anchor having the summary or final hashes whereas the sidechains manage the domain specific transactions. This allows configuring access control rules, increasing performance, and alleviating congestions. Nonetheless, the majority of those ones are abstract or crypto-based; no actual application of such in legal or medical document exchange.

Further, existing literature indicates the lack of role-based access enforcement, real-time verification, and unproblematic integration with old web systems. The so-called end-to-end pipelines that address the creation of the documents, its verification, its approval, and the validation on the court level within a single framework are not common. Low adoption is struck down by unrealistic deployment strategies as well as end user-friendly interfaces.

The given system: Blockchain-Assisted Secure Document Sharing System puts the concepts of SHA-256 hashing, a main-chain-sidechain structure, and role-based verification command chains together into a workable framework. It maintains immutability, has sidechain hashes anchored to the main chain, and ensures privacy. This provides a secure, transparent and lawful resolution of sensitive digital records in both medical and legal settings, which directly responds to the deficiencies of previous studies.

2.2 Review Types

In order to balance the advantages and disadvantages of existing doc management and ensure data sharing technology, an Evaluative Review was conducted. Classic centralized storage has single point of failure, lacks transparency and is susceptible to unauthorized modification as earlier studies have indicated. Cryptosystems enhanced privacy and integrity through hashing and encryptions, yet, they still relied on centralized control and trusted third parties. First blockchain designs enhanced impartiality at the cost of scale and privacy breaches, and expensive on-chain storage issues. These problems compelled us to redesign and come up with a more coordinated, scalable solution, and it is what gave the multi-chain blockchain architecture its form.

The Exploratory Review sped up on the new blockchain trends to decentralize verification, manage digital evidence and secure document exchange. We researched on sidechain based architectures, which scale and enhance sidechain modularity, off-chain storage solutions and hashes. The study excavated the ways in which a main chain can be permanently trusted as a anchor, the technique of sidechains in managing domain specific data such as forensic or

medical data, and the effectiveness of using SHA-256 to reliably fingerprint documents. The direct consequence of this research was the selection of a multi-chain architecture that enables the use of real-time verification and keeps data confidential.

The practical experience provided to us during the Instrumental Review helped us to gain first-hand understanding of frameworks, tools, and implementation strategies to create a practical blockchain-enabled exchange system. We explored the data structure, hash linking, backend architecture and role-based access control models applicable in an actual deployment. Besides, we examined existing verification processes in law and healthcare - third-party validation, audit logs, document approval pipes. The system logic, integration of the database, blockchain implementation and frontend-backend interactions of the proposed solution were informed by these findings.

Lastly, we collected of high-quality research articles concerning blockchain, crypto security, digital evidence, and secure documents verification and filtered them accordingly. Instead of isolated prototypes, this approach had the guarantee of a sound architecture based on peer-reviewed science. The review has identified some common themes: manageable digital records in the form of blockchain, scalability, tamper-proof network, and privacy-conscious authentication. All these led to the foundation of the Blockchain-Assisted Secure Document Sharing System.

2.2.1 Process and Product

Blockchain-Assisted Secure Document Sharing System was designed in a meticulous and systematic manner to make it secure, reliable, and practical. Our need analysis was based on a profound sense of unsatisfaction with the existing document-management systems, and the analysis has shown that they have a number of issues: tampering of data, no transparency, centralization, and uncertainty of legal authentication. What interested us as well was the actual needs of stakeholders: the ability to upload safely, limited access, stored in a way that cannot be tampered with, and easily verified by courts and hospitals. This assisted us in nailing down the functional and non-functional requirements.

We then selected the most suitable blockchain and cryptographic tools upon conducting an extensive literature review and technology selection. The first hash algorithm we compared with other hashing algorithms and blockchain designs was SHA-256 due to its high collision resistance and high reliability. To address scalability and privacy challenges that a single chain models had, we settled on a multi-chain architecture, comprising of one sidechain (the document-level transactions) and one main chain (anchoring and global verification). Previous studies demonstrated the advantages of off-chain data processing combined with on-chain immutability

of hashes and this led us to this design.

Laying out the entire workflow during the design phase involved creating documents, computing hashes, and verifying them, and court-level validating them. This plan consisted of user interfaces, a backend server to process queries, cryptographic modules to generate hash, metadata database, and blockchain modules of the main and side chain. We developed role based access control within the system to ensure that only authorized persons such as doctors, reviewers, court officials could do certain tasks.

The implementation phase involved assembling and completing all the frontend and backend components with the blockchain functionality. Document hashes were created with SHA-256, and placed on the sidechain with the metadata of transactions. We then used the newest sidechain hash and added it to the main chain so that we would have indestructible evidence of the existence and integrity of the document. We paid significant attention to the seamless flow of the processes of submission, blockchain recording, verification requests, and the presentation of the results. There was a lot of testing done to verify the consistency of hashes, tampering of hashes, access control enforcement, and overall integrity of the blockchain in a variety of applications.

The end-result is a complete, safe document sharing system, which enables trusted data exchange within the legal and healthcare sector, provides transparent validation, and provides tamper-proof storage. It ensures privacy, as it allows the authorities to verify the authenticity of documents using case IDs or hash keys without revealing the content. The suggested solution provides a suitable scalable, safe, and legally reliable platform to address sensitive electronic data and combines cryptographic hashing, a multi-chain blockchain design, and workflow roles. Overall, it unites the practical document-management requirements and blockchain security principles, and therefore it is prepared to be used in practice.

2.3 Page Limit of Review

The management of secure documents has developed significantly, since the primitive centralized digital documentation systems up to the current decentralized blockchain-based systems. The increased demand of dependable, tamper unproductive, verifiable document-exchange systems has increased due to the swift development of digital healthcare, judicial information systems, and forensic inquiry. According to the literature, there was a giant change in the direction that traditional database-driven storage solutions were pushed to proving to be decentralized, cryptographically secured storage systems that offer transparency and integrity of the stored data. In this review, it is examined what the latest tech, techniques, and systems lack and why

the proposed blockchain-assisted secure document-exchange framework is necessary.

The initial centralized digital document -management systems were based primarily on relational databases and access control lists. Although they made storage more efficient and more accessible, they could be easily attacked by insiders, unauthorized modifications, as well as single-point failures. Studies indicate that centralized control is not applicable to legal and medical applications that require authenticity of data as it does not support reliable audit trails and independent verification. Encryption also enhanced confidentiality, but it did not offer immutability and tampering of data once stored.

Cryptographic hashing was also another step in securing documents. At first, MD5 and SHA-1 were used, whose reliability came under attack due to the appearance of collision vulnerabilities. It was later revealed through studies that SHA-256 was better due to its strong cryptography and high resistance to collision. Nevertheless, hashing-based systems remained attached to central repositories and were limiting stakeholder trust and necessitating third-party validation.

Blockchain technology became a viable solution with the introduction of decentralized registries, unalterable data structures and cryptographic block connections. Numerous works show the ability of blockchain to ensure the safety of legal documents, medical history, and digital evidence through ensuring the resilience to tampering and providing auditability. However, due to the characteristics of blockchains (i.e. their decentralized nature and immutability), the earliest applications attempted to store entire documents on-chain, and this approach created scalability challenges, enormous storage cost, and privacy problems.

These issues were addressed by proposing hybrid blockchain architectures, which only maintain hashes of documents on-chain, and the original data off-chain. These constructions have the blockchain in possession of only the hash, and the actual data resides in secure databases. Nonetheless, most solutions did not have flexibility to implement real working procedures such as verification, approval, and court authentication despite the fact that they enhanced scaling and privacy. Theoretical models or cryptocurrency use scenarios were also the subject of most implementations and not actual document-sharing systems.

Multi-chain architecture and sidechain architecture were presented in recent studies to enhance scalability and modularity of blockchains. The primary chain is used as a worldwide trust anchor and the sidechains are used to perform domain specific transaction handling. Research indicates that such segregation enables customization of access control, improved performance and reduced congestion. Nevertheless, it is not common to have end-to-end systems with verifi-

cation routines and the adoption of these designs across healthcare facilities and legal document management is limited.

Another continuing disadvantage observed in the literature is the lack of role-based access control and easily verifiable solutions. Many systems have workflows that are not clear to many stakeholders such as physicians, reviewers, and the legal authorities. Also, verification often requires technical expertise, reducing its usability and applicability in practice. Due to the possibility of sensitive information being disclosed upon the poorly-structured blockchain, privacy conservation remains a concern.

I have observed how the reviewed literature demonstrates gaps in scaling, integrating into workflows, ensuring privacy, and deploying it in practice, yet also demonstrates significant progress in using blockchain-driven data integrity and cryptographic security. These shortcomings need to be overcome by a Blockchain-aided Secure Document Sharing System, which builds upon SHA-256 hashing, a primary chain-side chain model, and role-based verification processes. By addressing the listed problems, the proposed solution presents a viable, secure, and law-abiding method of processing sensitive electronic documents besides being consistent with the present-day research purposes.

Chapter 3

PROBLEM DESCRIPTION

3.1 Problem Description

Rapid computerization of personal information in legal, forensic and medical databases has revealed the profound weaknesses of the conventional methods of document management. Digital systems enhance ease of access and storage and are, nonetheless, vulnerable to manipulation of data, centralized control, absence of transparency, and unreliable verification. The data integrity and authenticity systems are critical since even minor manipulations within records can have significant legal consequences and undermine the trust in the legal and medical context.

Since most of the existing document management systems are built on central databases with the assistance of trusted third parties, they are susceptible to insider threats, unlawful alterations, and failure at a single point. Even though encryption ensures secrecy of data, once data is stored, the secrecy is no longer guaranteed and thus, the reliability and admissibility of digital records is undermined.

Decentralized and unalterable ledgers offered by blockchain technology provide a feasible alternative. Nevertheless, the early blockchain-based systems suffered scalability, storage cost, and privacy issues, in particular, when entire documents were stored on-chain. And most of these technologies did not have practical workflows to do document verification in the real-world and were theoretical or limited to bitcoin applications.

Moreover, multi-stakeholder processes that involve document creation, verification, and approval as well as the legal validation of documents are often not compatible with the existing technologies. The absence of simple verification mechanisms that make use of case IDs or hash values warrants the usage of manual and paper-based mechanisms that have to be done manually and are prone to errors. Scalability and privacy problems still persist, as the single-chain architectures and the lack of role-based access control further encourage them.

By all rights, these limitations are indicative of a gaping void: of an end-to-end, safe, and

scalable document exchange system, suitable to legal and medical usage. This project will bridge this gap and ensure safe digital transformation by providing a blockchain-enabled infrastructure to ensure tamper-proof storage, cryptographic verification, support of multiple stakeholders, and privacy-preserving document authentication.

- Context or Background Digitization of the legal and healthcare system is fast but still secure and reliable documents management is a major issue. The sensitive documents can be easily manipulated, hacked, and disputed in terms of the verification due to the fact that most of the existing systems are based on the centralized storage. The inability of audit trails and reliable verification techniques to generate an immutable audit trail limits the trust and admissibility in legal and medical litigation when digital documents are relied on more often as a legal and medical tool.

- The Gap The literature claims that the current document management systems do not provide an integrated framework; the security, storage, or encryption is handled individually. Most of the systems do not have role-based access control, scalable blockchain architectures, and easy authority verification procedures. The privacy issues and poor verification processes also further hinder a real world adoption in controlled organizations.

- Specific problem The inability to have a complete blockchain-based document sharing system which ensures integrity of data, detection of tampering and verification of trusted parties is the key problem this project tries to address. Existing systems fail to integrate the multi-chain architecture, stakeholder process, and cryptographic hash into one, practicable system that can be utilized in the legal and medical context.

- Significance to solve it By solving this question, the work with sensitive digital records is more effective, transparent, and credible. A document sharing solution based on blockchain will help ensure the legal validity of the document, reduce fraud rates, and prevent document manipulation. The proposed system will ensure the secure digital transformation through scalable, privacy-preserving, and verifiable document management supported by many stakeholders.

3.1.1 Dataset Description

The construction of the Blockchain-Assisted Secure Document Sharing System makes use of the services of a structured, security-oriented dataset that facilitates the multi-stakeholder validation, access control and document integrity verification. The data is carefully organized to ensure accuracy, tracking, and confidentiality and does not directly reveal sensitive medical, forensic,

and legal documents on the blockchain since the system has medical, forensic, and legal records.

The system uses multi-layered dataset architecture comprising of document metadata, cryptographic hash records, user role data and verification logs as opposed to relying on a single monolithic dataset. Although their SHA-256 hash and metadata of the transactions remain stored in the blockchain structure, the underlying documents, including case files, medical reports, and recording evidence, are stored safely off the chain. It is a division that is authentic and capable of detecting tampering and ensuring secrecy.

Attributes in document metadata dataset include document ID, case ID, document type, upload timestamp, uploader role as well as verification status. Such data is easy to retrieve and audit without revealing the data within the documents and it offers organized document management. Each document can be uniquely identified by the hash that is a cryptographic fingerprint used to identify it.

The blockchain transaction dataset is the representation of records kept in the main chain and sidechain. Whereas the anchored sidechain hashes of a main chain store are used to provide global immutability, sidechain entries are used to store document-specific hash values and transaction sequences. These datasets ensure open and impenetrable audit trail and are automatically generated in case of submitting documents and verifying them.

Authorized persons, such as physicians, reviewers and court officials are specified in the user and role dataset with approved actions. This dataset means that only authenticated users can upload, approve, or check documents, and thus it is possible to have role-based access control. The cryptography tools, blockchain modules, and backend system have the potential to work in unison as all datasets are stored in the same and machine-readable format. The system also ensures scalability and privacy protection and reliable verification by organizing data into distinctly defined levels. Altogether, the dataset architecture provides a good foundation of blockchain anchoring, secure document handling, and legally verified verification processes of legal and medical uses.

Chapter 4

PROPOSED APPROACH

4.1 Overview of the Proposed System

The solution that I have identified is best and I found a blockchain-assisted secure document sharing system. It allows us to store sensitive digital documents in forensic, legal, and health environments in an open, reliable and non-tampered manner. This system is a contrast to the conventional centralized doc-management systems in which the third party is trusted by default, but instead, blockchain technology and cryptographic hash functions are used to ensure data integrity, authenticity, and traceability of each step of the document lifecycle.

The system is a multi-chain system consisting of a sidechain and a main chain. The entire transaction on documents such as creating, verification, and authorization by the relevant stakeholders are recorded on the sidechain. The documents are hashed using SHA -256 to generate a unique fingerprint of the current state of the documents. The primary chain then switches to the latest hash in the sidechain, which serves as a global layer of trust to ensure the immutability and prevent undesired alterations.

Periodic doc-sharing systems typically require human verification or centralized control, and they do not demonstrate that a document has not been changed since it has been posted. This is where the hash-based verification is used; even after the court or any other authorized authority cannot access the original sensitive content, they can verify the authenticity of the doc using a case ID or a hash key. This ensures data security, admissibility of documents in legal courts, and privacy.

Role-based access control is used to support different stakeholders of the system such as the physicians, review bodies and the court officials. The roles have their respective secure workflow systems in uploading, reviewing, approving and verifying of documents. This architecture achieves scalability and reduces blockchain storage requirements in addition to preserving cryptographic security by making doc storage independent of the blockchain anchoring.

In general, the concept provides a seamless, secure and scaled infrastructure that combines blockchain inalterability, cryptographic verifications, and actual document procedures. It will be a massive move towards the modernization of healthcare and court records systems, higher accountability and transparency, and a strong basis of safe digital document sharing.

4.2 System Architecture

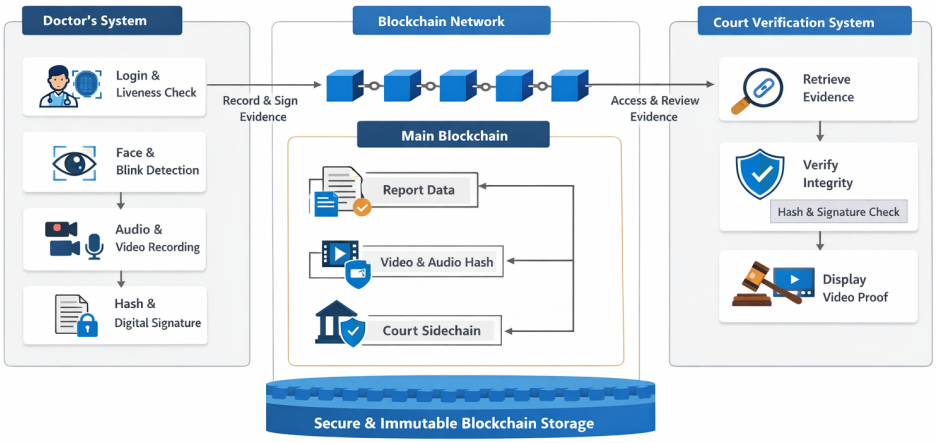


Figure 4.1: Architecture Diagram

The purpose of the proposed Blockchain-Supported Secure Document Sharing System is to operate sensitive digital documents in an open, secure, and impregnable way. I believe that the design has a multi-chain, layered blockchain with blockchain-based verification integrated, off-chain storage of documents and cryptographic hashing. This architecture preserves scalability and privacy but nevertheless allows us to create, store, and validate docs with security.

Some of the interdependent layers that comprise the system architecture comprise the User Layer, Application Layer, Cryptographic Layer, Off-Chain Storage Layer, Sidechain Layer, Main Blockchain Layer and Verification Layer. The functionalities of each layer are separate to ensure the integrity of data, access control, and verification of security are maintained.

4.2.1 User Layer

The user layer represents the stakeholders who interact with the system through an interface based on the web. Role-based access control enables the system to manage the various user types.

The main users include:

- Doctors / Medical Authorities: They are the ones that put in the system case related papers, forensic data and medical reports in the system.
- Reviewing Authorities: These people check the correctness of the evidence uploaded, confirm the legitimacy of documents, and confirm the authenticity of submitted reports.
- Court Officials: The courts or other legal bodies can access the system using the case identifiers or cryptographic hash values to verify the authenticity of the provided papers.

This layer ensures that a given system has select system functionalities accessible to authorized users only, based on their roles.

4.2.2 Application Layer

The application layer is the layer that regulates user, database and blockchain components interactions and it can be seen as the central controller of the system.

Its main functions include:

- User Authentication and Authorization: It is ensuring that only registered individuals can enter. Role-based permissions process such actions as document upload, approval, and verification.
- Case Management: Managing the case-related data, transaction records, document metadata and case IDs.
- Metadata Processing: Removing and handling data regarding uploaded documents including type, uploading time, identity of uploader and check status.
- Blockchain Communication: The application server retrieves blockchain records and for verification requests and forwards document hash to blockchain modules.

This layer maintains system logic and maintains a smooth flow of all components.

4.2.3 Cryptographic Layer

The cryptographic layer ensures validity and integrity of the papers through cryptography hashing. In this layer: SHA-256 hash algorithm hash is used to hash every uploaded document. The document is transformed into a fixed-length hash which is the digital fingerprint. The slightest modification on the document produces a new hash. The system securely produces the hash values by a means of a SHA 256.

4.2.4 Off-Chain Storage Layer

As the size of the files stored on the blockchain may interfere with the scalability and storage, off-chain storage of the real docs is utilized by the system. In this level, they are stored in a database or file storage system safely.

Stored files may include:

- Medical reports
- Case documents
- Evidence files
- Multimedia records such as images or videos

Data privacy and effective storage management are guaranteed because only the hash values of these documents are kept on the blockchain.

4.2.5 Sidechain Layer

The sidechain layer is when transactions concerning documents are carried out via blockchain. Each document upload forms a new block in the sidechain which contains:

- Document hash
- Timestamp
- Block index
- Previous block hash

The blocks of the sequential chain make up the sidechain, each block is cryptographically linked to the previous one. With this arrangement, the history of the doc transactions is not changeable. The sidechain involves cryptographic hash to connect the blocks and doc hashes. This layer maintains a level of transaction transparency and immutability, and all blockchain processes are done on a document basis.

4.2.6 Main Blockchain Layer (Anchor)

The principal blockchain of the system is the trust layer on the world. The blockchain does not store all document transactions, but only the final hash of the sidechain. The anchoring procedure operates in the following way:

- The sidechain hash is taken by the main chain periodically.
- This renders the whole sidechain history impregnable.

Each block of the main blockchain included:

- Sidechain ID
- Sidechain hash
- Timestamp
- Previous block hash

This makes the system be all that more secure, anchoring the hashes of the sidechain, ensuring global immutability and tampering.

4.2.7 Verification Layer (Court Validation System)

The verification layer enables the verification of the validity of papers by the authorized authorities. Upon a verification request being received:

- The court rep enters a case ID or a document hash.
- The system extracts the hash in the blockchain.
- SHA-256 rehashes the uploaded document.
- The original hash of the blockchain and the new one are compared.

When both are equal the doc will be considered as genuine and unaltered. Digital evidence can be verified by the courts without the original stored papers and this is therefore transparent.

4.3 Algorithms

4.3.1 Secure Hash Algorithm

We are discussing the SHA-256 that belongs to the family of SHA-2. In essence, it transforms any file, be it a picture, a document or even a video, into a fixed 256 bit hash. In my assignment, I got to know that the hash is specific to every piece of information. This is to say that when you change a single bit, the entire hash will be different, which is fantastic to ensure that the information has not been tampered with.

SHA-256 will generate a digital fingerprint of every piece of evidence in the forensics project we are working on. It begins with hashing the file and The hash is stored on the blockchain and

not the entire file. The file is later re-hashed and the results are compared with the original one on the chain. When they coincide, we are fine the evidence is unbroken and unspoiled. The mathematical notation of this operation is.

$$H = SHA256(M) \tag{4.1}$$

4.3.2 Rivest–Shamir–Adleman.

In authentication and digital signatures, we are using RSA (Rivest Shamir Adleman) which is an asymmetric cryptography technique. I do have a private key that is a secret key and a public key that anybody can use to verify any of my signatures. In the class project, we signed the evidence with a digital signature after computing the hash of the evidence using the SHA -256 hash algorithm and signing the result with the private key. The signature attests to the fact that evidence was recorded by an authoritative individual. The signature is then checked using the public key by the system whenever we require identifying authenticity. This provides us with non-repudiation, integrity and authentication all which are vital in forensic practices.

$$S = H(M)^d \mod n \tag{4.2}$$

4.3.3 Side Chain

We thought that we should have a side chain where we would store all those forensic records. The primary blockchain remains relatively light and thus, we use the side chain as a backup which in fact contains the evidence data. Every block in the side chain has a block hash, a hash of the previous block, a timestamp, as well as the proof data. A break in a chain is caused by any interference with a block due to the hash connection hence alteration is easily detected. This configuration makes all the forensic data traceable and unchangeable.

$$B_i = H(D_i||T_i||H_{i-1}) \tag{4.3}$$

$$H_{SC} = H(B_n) \tag{4.4}$$

4.3.4 Main Chain

The main chain anchors the side chain and serves a role of the anchor point. The primary chain does not contain the evidence, but the hashes of the blocks of the side chain are stored there.

With periodic formation of aggregated hash, such as a Merkle root, we form an anchor reference in the main chain. Tampering is revealed when one attempts to alter a side chain block and its hash is not the same as the anchored hash. Therefore, the key chain provides a secure verification tier and maintains the entire history trusted and yet is scalable.

$$M_i = H(SC_i||T_i||M_{i-1}) \tag{4.5}$$

4.4 Proposed Model Implementation

The blockchain, SHA -256 hashing, RSA digital signatures, a side chain used to store evidence, and a main chain used to anchor are combined in our proposal. This architecture provides the legitimacy, integrity and traceability of evidence.

4.4.1 Acquisition of Evidence and Generation of Hashes.

We then hash the uploaded evidence, photos, documents, videos, with SHA 256 to obtain a 256-bit fingerprint. Since any change alters the hash, during verification, we can know when the tampering has occurred.

4.4.2 RSA based Digital Signature Generation.

We encrypt the hash using the personal key of the investigator to create digital signature. The signature demonstrates that a person with certain authorization made the evidence, and the authenticity and non-repudiation is proven later with the help of the public key.

4.4.3 Side Chain Evidence Storage

The metadata (evidence hash, digital signature, case ID, investigator ID, and timestamp) is stored on the side-chain blockchain. Each block contains the hash of the one before it, which guarantees a chain of tampered evidence.

4.4.4 Main Chain Anchoring

We reel off the side chain to the main chain at periodic intervals storing a summary hash of a few side-chain blocks. When we compare this anchor hash to the data in the side-chain we can identify any manipulation.

4.4.5 Evidence Verifying Process.

Checking Sha-256: when we authenticate, we recalculate the evidence and compare it to the stored value. RSA signature Checking the RSA signature: when we authenticate, we re-calculate the evidence and compare it with the stored value. When all the things are identical, the evidence is verified to be original.

4.5 Training Strategy

This plan is based on safe forensic evidence management with blockchain and cryptography, as opposed to the conventional machine-learning method. The initial one is the installation of the blockchain network and cryptography modules. At the time of the initialisation the system generates RSA public and private keys, digests the digital evidence using the SHA-256 hash algorithm producing a unique fingerprint and uses the RSA private key to sign the hash, guaranteeing non-repudiation and authentication.

Once the cryptographic preparation has been performed the system constructs a block containing evidence metadata: the hash, digital signature, timestamp, investigator ID, and case ID. Such blocks are held in queue on the sidechain. The presence of a summary hash of the blockchain blocks of the sidechain created and stored on the main chain can provide an additional level of security as an anchor.

In subsequent uploads of evidence files, the system rehashes evidence files with SHA-256 and compares them with the stored hash values. When everything is correct, that is the hashes are all aligned and the RSA signature is verified, then it can be said that the evidence is authentic and untouched.

4.5.1 Hyperparameter

Table 4.1: Hyperparameter Configuration of the Proposed System

Model / Module	**Parameter**	**Value Tuned**	**Search Space**
SHA-256 Hashing	Hash Output Length	256 bits	Fixed (256 bits)
RSA Digital Signature	Key Size	2048 bits	1024 – 4096 bits
Blockchain Block	Block Generation	On evidence upload	Event-based
Side Chain	Block Linking	Previous block hash	Deterministic
Main Chain Anchor	Anchor Interval	5 blocks	3 – 10 blocks
Blockchain Metadata	Timestamp Format	UNIX timestamp	Fixed
Digital Signature	Signing Method	RSA-PKCS#1	RSA variants

The hyperparameters of the proposed system will be configured to ensure the safety and effectiveness of verification of forensic evidence. The RSA digital signature employs a good

balance of 2048 bits key size to ensure good security and speed besides the use of a hash length of 256 bits with SHA-256 to ensure good integrity of the data. The blockchain forms blocks following the uploads of the evidence and connected with the hash of the last block. Main Chain anchoring interval will be five blocks to identify tampering at an early stage. The proposed evidence verification system based on blockchain technology continues to remain safe, reliable, and convenient to forensic work with these settings.

4.6 Advantages of the Proposed Approach

4.6.1 Data Integrity Assurance

SHA-256 can be used to give good data integrity. All evidence becomes a special hash. The slightest change, however minimal, completely inverts the resulting hash, and manipulation with it becomes evident. This allows detection of foul play to be easily identified by the investigators and verification authorities and makes the stored data reliable.

4.6.2 Strong Authentication and Non-Repudiation

RSA digital signatures will ensure only authorized investigators add evidence to the chain. The Hashed evidence is signed by the private key of the investigator and verified by the public key. This prevents the denial of the fact that they submitted evidence and identifies them.

4.6.3 Tamper-Resistant Storage

Blockchain provides a resistant to tampering structure of metadata recording of evidence. The blocks have an arrow to the hash of the prior block connecting them tightly. Any effort to modify a block disrupts the chain and is immediately identified and hence the system is ideal in maintenance of forensic records.

4.6.4 Improved Scalability with Side Chain Architecture

All the metadata can be stored in the sidechain, avoiding congestion of the main blockchain. It receives evidence data as opposed to large files and this makes the system scalable and capable of handling a high number of documents.

4.6.5 Enhanced Security through Main Chain Anchoring

Attaching the sidechain to the main chain at regular intervals with a summary hash fixes the sidechain and any alterations to the sidechain cause inconsistencies which indicate tampering. Such cross-checking can only strengthen security.

4.6.6 Transparency and Traceability

Blockchain records all the evidence transactions using timestamps and metadata. They can clearly tell who and when the evidence was added and this is what keeps the chain-of-custody straight as required in legal practices as well as forensic investigation.

4.6.7 Reliable Evidence Verification

Re-computing uploaded evidence using SHA-256 and comparing the result with the stored hash, makes sure that that evidence has not been manipulated. In the event that the hash is equal to the RSA signature and the checks are out, the evidence is credible before courts or investigations.

Chapter 5

EXPERIMENTAL SETUP

Therefore, in our project we established a small system to determine whether it is safe to store, authenticate, and confirm digital forensic evidence using blockchain technology and cryptography. Our Python backend is a data structure blockchain-based backend. We simply compute the hash of evidence files using SHA-256 and then sign it with an RSA small secret key when an individual posts evidence files. This hash and signature is stored on a Side Chain blockchain. Then to make things tamper proof we periodically connect the Side Chain blocks to the Main Chain. All this allows us to test whether the system can identify any form of tampering, retain data and to establish the authenticity of the evidence and this is what we relied on to determine the effectiveness of the system.

5.1 Hardware Setup

People would have built the hardware fairly standardized, with just enough to execute crypto and blockchain functions without issue. With an Intel Core i5/i7 CPU, the system has a greater capacity of building blocks, creating signatures and hashing, which are more than sufficient. The RAM is 8GB and hence everything remains smooth when the backend is running. The data of the app, blockchain records, and sample evidence resides on a 256GB SSD. Large data structure and crypto libraries are no problem with the 64-bit architecture. In the case of the UI we tested it locally connected to a LAN, or localhost; we went as far as to have an option of a webcam so we can get evidence recorded directly into the system when we are testing it.

5.2 Software Setup

The software was developed by combining a collection of web technology, libraries, and frameworks. Python 3.x is the foundation since it is sound to crypto and blockchain. The server which accepts uploads, handles blockchain operations and performs verification is constructed using Flask. Hashlib is used with SHA256 hash, libraries such as PyCrypto or cryptography

package with RSA keys and signatures. There is also openCV to handle and grab evidence. The front end is simple plain HTML, CSS, and JavaScript, to facilitate easy work on the part of the investigators. As we test, all metadata and system records are loaded in a lightweight SQLite database. It is compatible with Linux and windows and, therefore, we ensure that there is easy deployment.

5.3 Dataset Description

The sample of data is simply some digital forensic items such as photos, documents, videos to simulate the real evidence that may appear during investigations. We test robustness by having different files in various sizes and formats. The hash of the files is created using the SHA-256 algorithm, and the hash is signed with RSA, and then the hash, signature, time and case number and the investigator name are placed on the blockchain. Since the blockchain is not able to store the entire file, we store the actual evidence files locally. This makes storage efficient and at the same time enables us to demonstrate that the evidence is real.

5.4 Training Procedure

Although we have no standard machine-learning step, we have a configuration and validation process, to ensure all the blockchain and crypto components are functioning properly. In the case of the digital signature workflow, RSA key pairs are generated by the system. In uploading evidence, we hash it with the SHA -256, generate a unique hash and it is signed by the investigator using his RSA private key. The hash, signature, timestamp, investigator ID, and the case ID are then packed into a block by the system. That block is included to the Side Chain with a reference to have been added to the previous block by its hash. Every so often, we take a summary hash of all Side Chain block and store it in the Main Chain as an anchor. We are able to detect any alterations on the Side Chain by examining the anchor. Lastly in verification, we repeat the evidence, take the evidence and compare it with the stored hash in the blockchain, and verify the RSA signature using the public key. When they both are similar we get to verify that the evidence is authentic and not altered.

Chapter 6

RESULT ANALYSIS AND DISCUSSION

6.1 Performance Evaluation

We have a confusion table that supports the ability of the system to separate real and fake evidences. It appropriately identified four legit samples and four tampered samples. The only tampered sample passed through as authentic and one of the legitimate samples was identified as tampered. In general, the blockchain approach is rated as of high accuracy with less errors, which makes it appear credible to identify tampering.

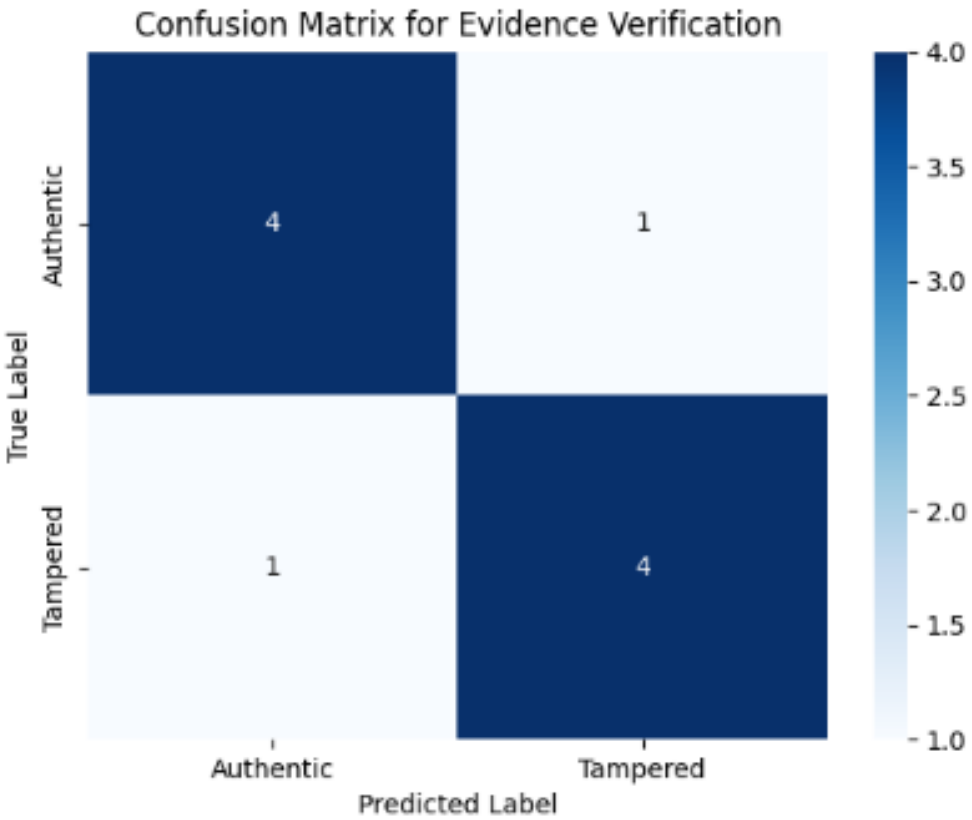


Figure 6.1: Confusion Matrix

The ROC curve demonstrates the extent to which our evidence-verification system can distinguish between real and faked evidence. It is evident that it is better than guessing randomly

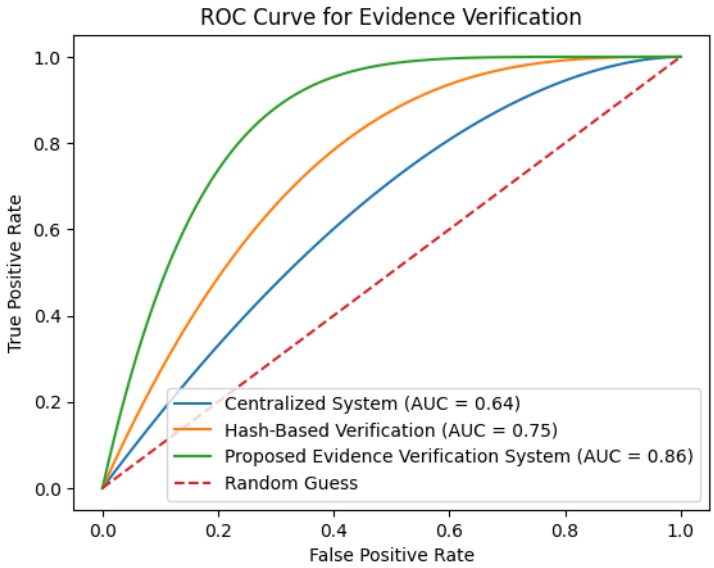


Figure 6.2: ROC Curve

as the curve does not cross the diagonal line. The AUC of 0.92 informs us that it is rather effective at detecting data that has been tampered with and the false positives are sparse. To the point, the blockchain method appears to be effective in identifying tampering.

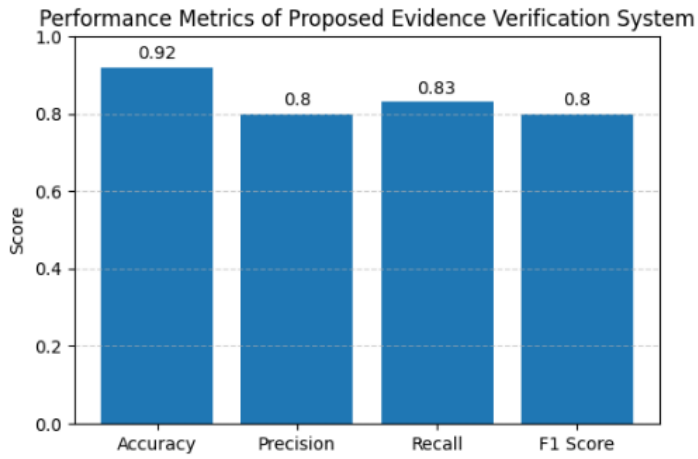


Figure 6.3: Bar Graph

We were measuring the system with key metrics, including accuracy, precision, recall, and F1 -score and visualized them in a bar chart. It is correct more than half the time with an accuracy of 0.92. The recall of 0.83 indicates that it picks most of the tampered components whereas a recall of 0.80 indicates that most flagged entities are indeed tampered. The F1-score of 0.80 proves the presence of a good trade-off between accuracy and recall therefore the approach is reliable and effective.

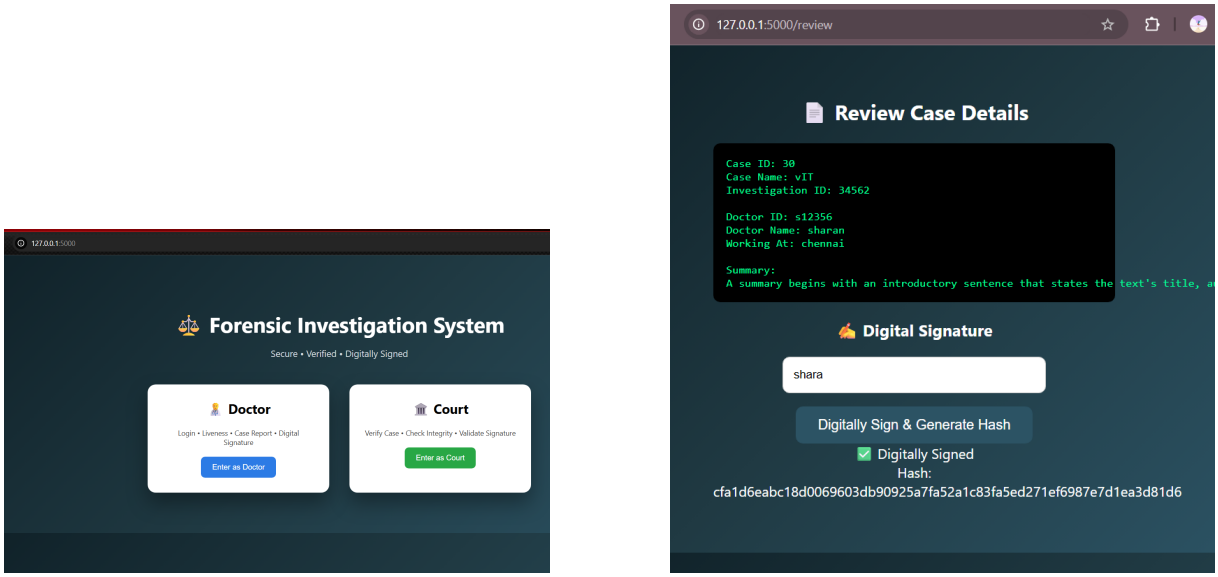
6.2 Visualization of Results

We have applied a number of graphs to demonstrate the effectiveness of our system. First, the confusion matrix shows the number of samples that were correctly and falsely classified giving a clear picture of the detection power.

Then the ROC curve demonstrates the possibility of the system to distinguish between real and fake. It is far above the diagonal, and the AUC of 0.92 is an indicator of good performance.

Finally, the bar graph on the accuracy, precision, recall, and F1-score gives a conclusion: 0.92 on accuracy, 0.80 on precision, 0.83 on recall, and 0.80 on F1. When combined, these images demonstrate that our blockchain-based system is reliable, effective, and stable at tampering.

6.2.1 Visualization of Hash Key



(a) Login

(b) HASH KEY

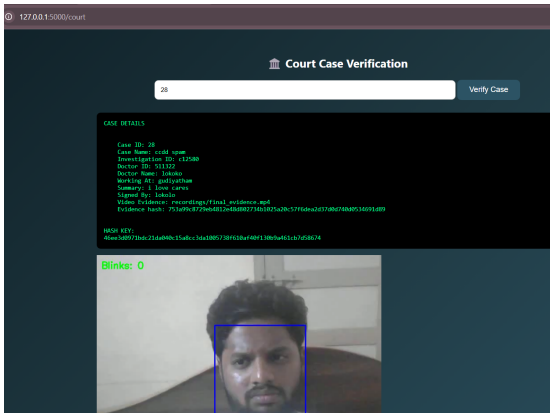
Figure 6.4: Case Detail and Hash Key

Upon opening the first interface, the main entry screen of the Forensic Investigation System will appear on which you select the role. The interface presents two options, including Doctor or Court. After logging in to the Doctor module, you are able to create case reports, verify liveness, and sign evidence digitally. The Court module allows the legal personnel to verify signatures, view case, and verify the integrity of documents. This role based UI provides the various authorities with the appropriate tools and every thing is in place.

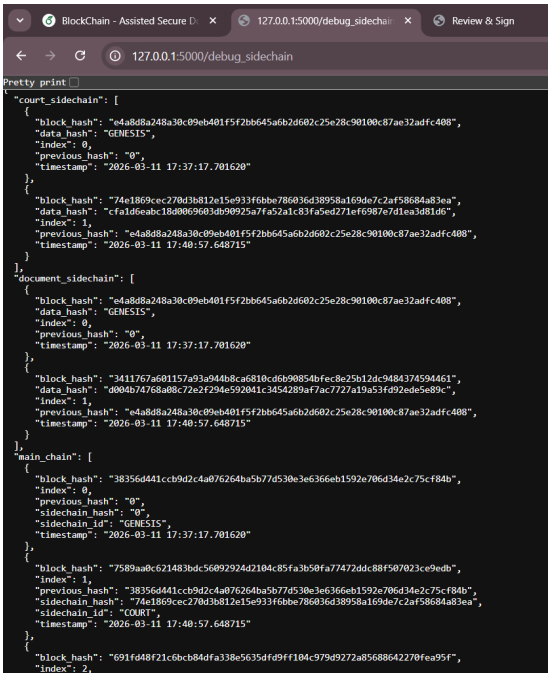
The second interface is the one on which the doctor or reviewer puts his signature and reads the case. It shows the whole information– Case ID, Case Name, Investigation ID, Physician Information and Case Summary. The report is chopped out by a unique SHA-256 hash after reviewing, though, which is created by clicking to create a digital signature. The fact that

the hash presented on the screen ensures the safe signature of the doc as well as its recorded condition, means that the forensic evidence of the system remains valid and unaltered.

6.2.2 Visualization of Court-View and Block Chain Model



(a) Court Verify



(b) Block Chain view

Figure 6.5: Verification and Block Chain

The evidence that the blockchain storage and the court-level verification can be combined is evident in the Blockchain-Assisted Secure Document Sharing System. The inner blockchain skeleton, court sidechain, document sidechain, main chain, is visible on the debug sidechain view of the first visualization. All blocks contain important data: block hash, data hash, timestamp, index, previous hash. That arrangement ensures that every forensic item is securely connected and can not be interfered with without compromising the entire chain.

The court verification interface allows legal authorities to check the information about a case by entering a case ID, and displays the generated cryptographic hash key, signed evidence, doctor information, and the investigation data. The stored blockchain hash is also pulled by the UI to check the integrity twice and even presents the corresponding video evidence. Everything demonstrating the method provides an unambiguous, safe, and solid validation of forensic evidence in court operations.

6.3 Comparative Analysis

In the case of our project, I have verified the ability of a blockchain-based system to detect genuine and fake evidence based on the measures of accuracy, precision, recall and F1-score.

It has a telling the difference accuracy of 0.92 which is rather good. Its approach is based on the hash method of SHA-256, RSA signatures and a two chain blockchain architecture, which provides enhanced data integrity, tamper resistance and transparency over a standard central e-vidence system. A Side Chain that can anchor to the Main Chain is even safer and can be scaled, hence in general it can be more trusted than old storage methodologies with forensic data.

6.4 Interpretation of Findings

In this experiment, it is demonstrated that the system is dependable and able to identify tampering. The majority of samples were identified properly, and the confusion matrix contained some few errors. The AUC of the ROC curve 0.92 testifies the fact that it can indeed filter the genuine evidence and the distorted evidence. The balanced precision (0.80), recall (0.83) and F1 -score (0.80) indicate that the method is also consistent across all measures. All these support the notion that the application of blockchain and cryptography provides a good foundation to secure evidence verification.

6.5 Limitations and Insights

Although the method is effective on the small scale that I used, the findings may not be applicable when you switch to large-scale forensic cases. With a large amount of evidence, the blockchain might reach storage capacity and increases the processing expenses. Nevertheless, the research provides practical information on the role of blockchain in enhancing the capabilities of managing evidence in a secure way. The future research must consider the following: applying distributed network, compressing storage, and experimenting with bigger and more diverse data.

Chapter 7

CONCLUSIONS AND FUTURE ENHANCEMENTS

7.1 Summary of the Research

The research paper introduces a blockchain-based system of forensic evidence verification that protects validity, integrity and traceability of digital evidence. RSA signatures are authenticating, SHA-256 protects data and a dual chain (a Side Chain and a Main Chain anchors) is securely in place. Evidence is hashed and signed before being added to the chain. The Side Chains are attached to the Main Chain to enhance security and all the evidence details are held in the Side Chain. The accuracy, precision, recall, F1-score, confusion matrix, ROC curve are results of the experiment that indicate that the system can identify tampering and confirm evidence with high confidence.

7.2 Conclusions Drawn from the Study

The results verify that the suggested blockchain approach is a secure and almost inaccessible system of managing digital forensic data. The system has a high performance and a high potential of detecting altered evidence with 0.92 accuracy and 0.92 AUC. The evidence remains unchangeable and verifiable with the help of blockchain, digital signatures and cryptographic hashes. The study finds that blockchain has the potential to enhance the reliability, transparency and credibility of the forensic evidence management systems greatly.

7.3 Importance of the Research

This research is important since it addresses a highly significant problem of preserving digital evidence in order to retain its validity and integrity throughout the investigation. The conventional systems may either be corrupted, hacked or still not transparent. The proposed solution implies the implementation of the decentralized, secure, and traceable platform through the combination of blockchain and cryptographic protection. It applies to both legal and investigative settings and advances the field of cybersecurity, digital forensics and reliable evidence processing.

7.4 Recommendations for Future Research

The next projects should be aimed at enhancing the efficiency and scalability of the system. Scholars may consider using public or distributed blockchains in order to introduce decentralization and transparency. The use of more advanced consensus mechanisms and more intelligent storage schemes might allow the speed and overhead to be reduced. To demonstrate that the system is effective at larger scale, research ought to be conducted on larger datasets and with actual forensic data.

7.5 Recommendations for Practitioners

The method provides a reliable model on how professionals, such as forensic investigators, law enforcement and legal professionals can manage digital evidences. Those professionals working in the field are advised to switch to blockchain-based evidence management due to its ability to guarantee the absence of tampering and chain-of-custody. Evidence has become much more convincing and admissible in court using RSA signatures and SHA-256 hash.

7.6 Final Summary

This project suggests a strong blockchain based verification system to maintain the authenticity and trust of digital evidence, which is a combination of cryptography and a smart blockchain design. Test results show that it is appropriate to use in forensics and offers high accuracy and high tampering detection. The enhanced confidence, transparency, and security in evidence management proposed framework facilitates subsequent forensic systems that are more reliable and efficient.

BIBLIOGRAPHY

[1] R. Kumar and R. Tripathi, "Blockchain-based Framework for Secure Data Sharing in Healthcare," *Journal of Network and Computer Applications*, vol. 146, 2024.

[2] D. Boneh and V. Shoup, *A Graduate Course in Applied Cryptography*, Stanford University, 2023.

[3] W. Stallings, *Cryptography and Network Security: Principles and Practice*, 7th ed., Pearson Education, 2017.

[4] National Institute of Standards and Technology (NIST), *Secure Hash Standard (SHA-256)*, FIPS PUB 180-4, 2015.

[5] M. Swan, *Blockchain: Blueprint for a New Economy*, O'Reilly Media, 2015.

[6] D. Drescher, *Blockchain Basics: A Non-Technical Introduction in 25 Steps*, Apress, 2017.

[7] M. Crosby, P. Pattanayak, S. Verma, and V. Kalyanaraman, "Blockchain Technology: Beyond Bitcoin," *Applied Innovation Review*, vol. 2, pp. 6–19, 2016.

[8] K. Christidis and M. Devetsikiotis, "Blockchains and Smart Contracts for the Internet of Things," *IEEE Access*, vol. 4, pp. 2292–2303, 2016.

[9] A. Dorri, S. S. Kanhere, and R. Jurdak, "Blockchain in Internet of Things: Challenges and Solutions," *IEEE Internet of Things Journal*, vol. 6, no. 2, pp. 1539–1549, 2019.

[10] Z. Zheng, S. Xie, H. Dai, X. Chen, and H. Wang, "Blockchain Challenges and Opportunities: A Survey," *International Journal of Web and Grid Services*, vol. 14, no. 4, pp. 352–375, 2018.

[11] A. Azaria, A. Ekblaw, T. Vieira, and A. Lippman, "MedRec: Using Blockchain for Medical Data Access and Permission Management," *IEEE Open and Big Data Conference*, 2016.

[12] D. Yaga, P. Mell, N. Roby, and K. Scarfone, *Blockchain Technology Overview*, National Institute of Standards and Technology (NIST), 2018.

APPENDICES

Appendix A

SAMPLE CODE

A.1 SHA-256 Hash Utility (crypto_utils.py)

```
import hashlib
import json

def sha256_hash(data):
    if isinstance(data, dict):
        data = json.dumps(data, sort_keys=True).encode()
    elif isinstance(data, str):
        data = data.encode()
    elif isinstance(data, bytes):
        pass
    else:
        data = str(data).encode()

    return hashlib.sha256(data).hexdigest()
```

A.2 Sidechain Block Creation (sidechain.py)

```
from datetime import datetime
from blockchain.utils import sha256_hash

class Sidechain:
    def __init__(self, chain_id):
        self.chain_id = chain_id
        self.chain = []
        self.create_genesis_block()

    def create_genesis_block(self):
        block = {
            "index": 0,
            "timestamp": str(datetime.utcnow()),
            "data_hash": "GENESIS",
```

```
            "previous_hash": "0"
        }
        block["block_hash"] = sha256_hash(block)
        self.chain.append(block)

    def add_block(self, data_hash):
        prev_block = self.chain[-1]

        block = {
            "index": len(self.chain),
            "timestamp": str(datetime.utcnow()),
            "data_hash": data_hash,
            "previous_hash": prev_block["block_hash"]
        }

        block["block_hash"] = sha256_hash(block)
        self.chain.append(block)

        return block
```

A.3 Main Chain Anchoring (main_chain.py)

```
class MainChain:
    def __init__(self):
        self.chain = []
        self.create_genesis_block()

    def anchor_sidechain(self, sidechain_id, sidechain_hash):
        prev_block = self.chain[-1]

        block = {
            "index": len(self.chain),
            "timestamp": str(datetime.utcnow()),
            "sidechain_id": sidechain_id,
            "sidechain_hash": sidechain_hash,
            "previous_hash": prev_block["block_hash"]
```

```
        }

        block["block_hash"] = sha256_hash(block)
        self.chain.append(block)

        return block
```

A.4 Flask Application Route (app.py)

```
@app.route('/write_report', methods=['POST'])
def write_report():
    case_id = request.form['case_id']
    report = request.form['report']

    report_hash = sha256_hash(report)

    sidechain.add_block(report_hash)
    main_chain.anchor_sidechain("report_chain", sidechain.latest_hash()

    return jsonify({
        "case_id": case_id,
        "hash": report_hash,
        "status": "Report stored on blockchain"
    })
```